

Cumulative distribution function (CDF) is defined as

$$\textcircled{*} \quad \underline{F(t) = P(X \leq t)} \quad \text{for} \quad \underline{-\infty \leq t < +\infty}$$

(any real t)

See p. 88 of the textbook for definition
(also in the continuous case) and the property

$$F'(x) = f(x)$$

where $f(x)$ is PDF
of a random variable.

See also § 3.2 of Bertsekas & Tsitsiklis book
if you have it.

- In the homework you are assumed to compute CDF by definition $\textcircled{*}$
- Another way to compute CDF's (which works only in continuous case) is from PDF

$$F(x) = \int_{-\infty}^x f(t) dt.$$

Example 1 CDF of Bernoulli random variable:

$X = 1$ with prob $1/2$
 0 with prob $1/2$